

KnockKnock

Low-Cost IoT Intrusion Detection for Doors and Windows

IoT: Algorithms and Services

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The Problem in 10 Seconds

1 Problem

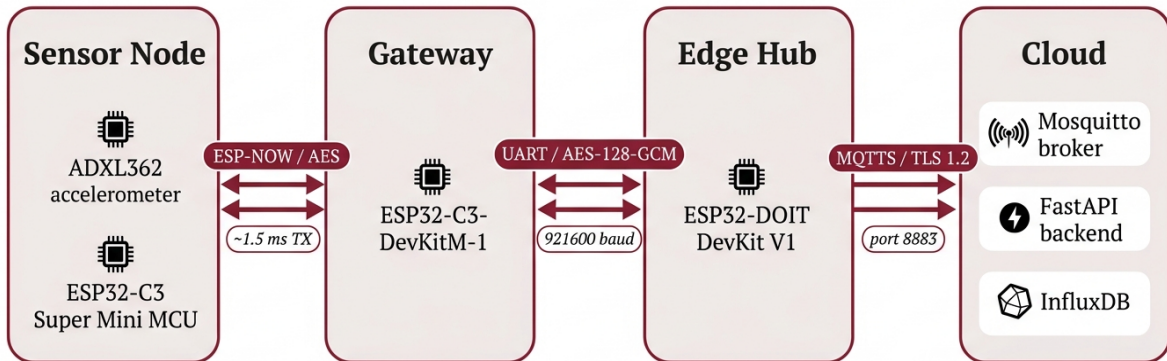
Detect intrusions through windows, without wiring, for months on a battery.

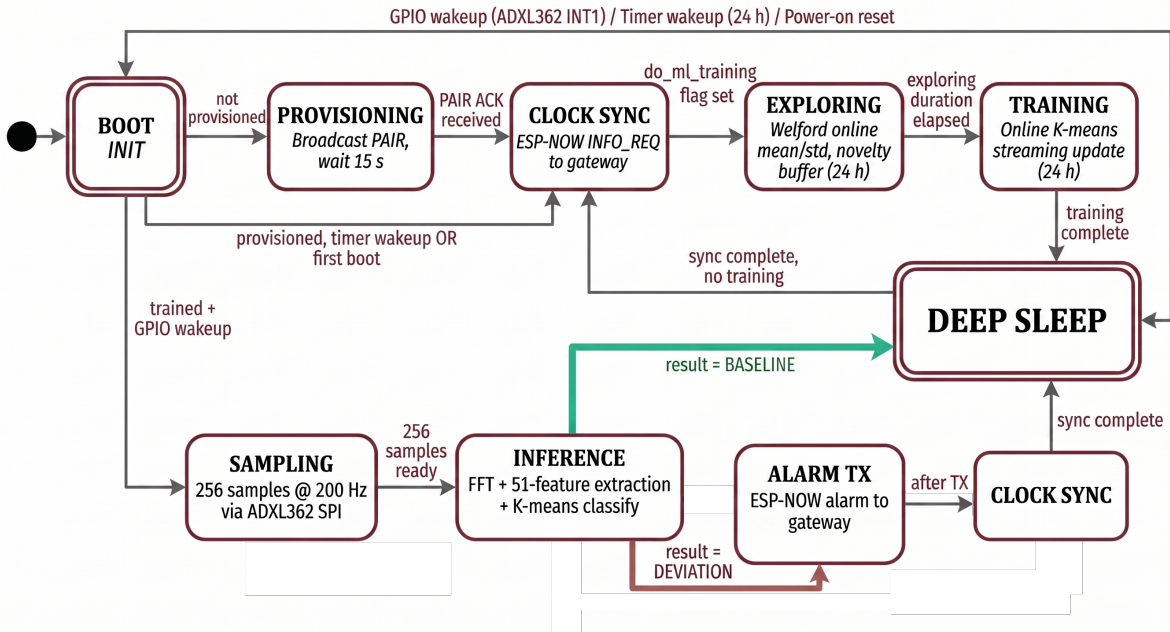
Constraints

- No wiring (renters, retrofit installs)
- Battery must last months, not days
- Sub-€10 per sensor node
- Must detect vibrations accurately from inside the window frame

Requirements

- Quiescent current $< 100 \mu\text{A}$
- Detection latency $< 5 \text{ s}$
- Battery life $> 6 \text{ months}$
- False positive rate near zero (traffic, wind, rain unacceptable)







System Requirements

3 Requirements

Requirement	Target	Achieved	Mechanism
Quiescent current	$< 100 \mu\text{A}$	$\approx 50 \mu\text{A}$	ESP32-C3 deep sleep + ADXL362
Active current	$< 25 \text{ mA}$	$\approx 20 \text{ mA}$	ESP32-C3 RISC-V
Detection latency	$< 5 \text{ s}$	$\approx 2.7 \text{ s}$	HW interrupt + FFT
Battery life	$> 6 \text{ months}$	$\approx 8.7 \text{ months}$	Duty cycle + ADXL362
Wireless range	$> 10 \text{ m}$	$\approx 50 \text{ m}$	ESP-NOW 2.4 GHz

Quiescent mode (nothing happens):

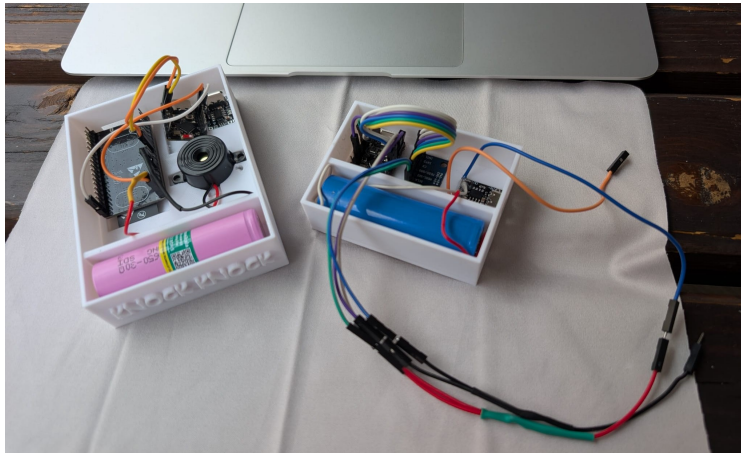
$$I_q = \underbrace{50 \mu\text{A}}_{\text{MCU sleep}} + \underbrace{270 \text{ nA}}_{\text{ADXL362 activity detect}} \approx 50 \mu\text{A}$$

MCU wakes *only* when ADXL362 INT1 fires; the system is off by default.



Our Prototype

4 Sensor Node





Wake-up Sensor: LM393 → ADXL362

4 Sensor Node

Before

LM393 comparator
Always-on: 1.5 mA
Piezo-based, no data
Manual threshold (trimmer)

After

ADXL362 accelerometer
Activity detect: 270 nA
INT1 → GPIO wakeup
5500× less current

- **SW-configurable threshold:** activity level set via SPI register; no physical potentiometer needed on re-deployment
- **200 Hz sampling:** window vibrations are sub-100 Hz
- **Internal mount:** avoids weather exposure; tunable SW threshold compensates for reduced sensitivity

Net effect

Threshold reconfiguration: SPI register write at boot; no physical enclosure access required.
HW threshold (150 mg default) blocks traffic, HVAC, footsteps: MCU never wakes on ambient vibrations.



On-Device K-Means++ Pipeline

4 Sensor Node

1. Training

- 512-pt FFT $\rightarrow$ 51-D feature vector
- **Online K-Means++:** K-Means++ init + incremental online updates
- 24h exploration phase on actual window
- No full dataset retained on device

2. Threshold

- Per-cluster: $\theta_k = \mu_k + 3\sigma_k$
- Computed from training distribution
- $K = 10$ clusters
- Self-calibrates without retraining from scratch

3. Inference

- Compute distance d^* to nearest centroid
- $d^* \leq \theta_k$: **baseline** (ignore)
- $d^* > \theta_k$: **alarm**
- $O(K \cdot d)$: 510 MACs at $K=10$, $d=51$; < 10 ms on ESP32-C3

Evaluation

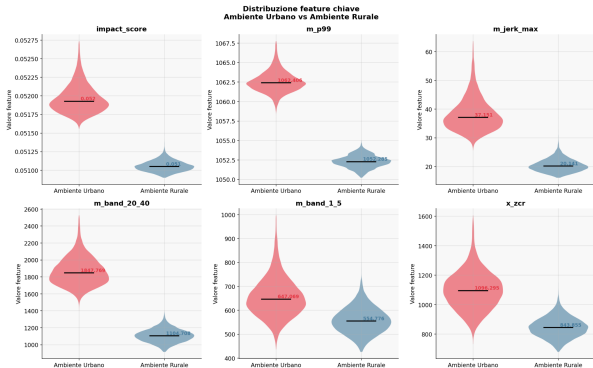
Tested across multiple installation positions on the same window frame; θ_k updated incrementally without full retraining.



Why On-Device Learning?

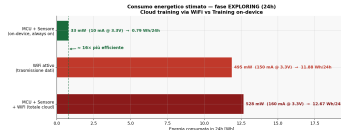
4 Sensor Node

Environment heterogeneity



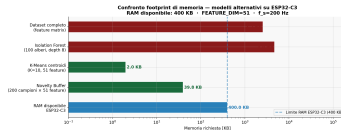
Urban (double-pane) and Rural (single-pane) windows produce structurally different PSDs; a pre-trained cloud model does not transfer between environments.

WiFi energy overhead $\times 16$



EXPLORING (24 h): on-device 0.79 Wh vs. WiFi 12.67 Wh; 16 $\times$ overhead incompatible with 3500 mAh energy budget.

Isolation Forest exceeds device RAM



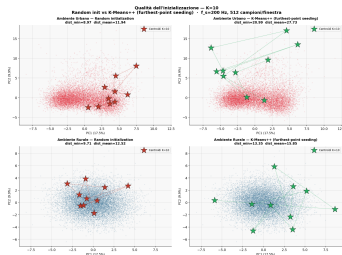
Isolation Forest: 4.7 MB; K-Means + novelty buffer: 42 KB; ESP32-C3 available: 400 KB.



K-Means++ Seeding, Novelty Buffer, and K Selection

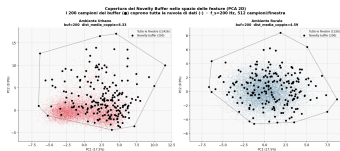
4 Sensor Node

K-Means++ seeding quality



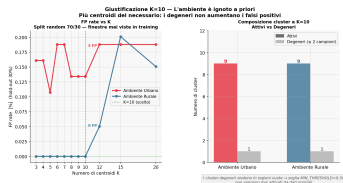
Farthest-point init: Urban $\bar{d}$ 11.94 $\rightarrow$ 27.73,
Rural 9.71 $\rightarrow$ 15.85; $2.3\times$ improvement in
inter-centroid spread prevents early collapse.

Novelty buffer coverage (PCA 2D)



200 retained samples (highest centroid
distance) cover the full 12,426-window cloud;
 $200 \times 51 \times 4 \text{ B} = 40.8 \text{ KB}$ on-device.

K = 10: FP sweep and cluster composition



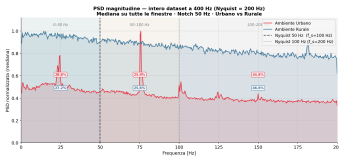
FP-rate minimum at $K = 10$ for both
environments; 1 degenerate centroid per
environment, FP rate unaffected.



Sampling Rate, Voting Strategy, and False Positive Reduction

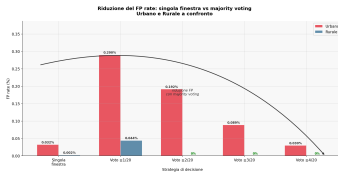
4 Sensor Node

PSD justifies 200 Hz + band 40–100 Hz



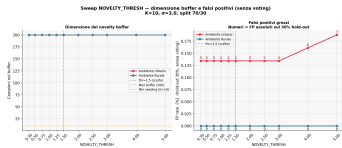
Vibration peak at ~ 75 Hz (Urban); 50 Hz mains interference. Nyquist at 200 Hz covers 40–100 Hz band; higher rates add no discriminative content.

Majority voting reduces FP to 0%



Single-window: Urban 0.032%, Rural 0.002%. All Urban FPs are isolated; MIN_CONSECUTIVE = 2 (5.12 s) reduces FP to 0% on both environments.

NOVELTY_THRESH robust up to 3.0



Buffer saturates at 200 samples regardless of threshold. FP rate flat up to threshold 3.0; operating point 1.5 provides $2 \times$ headroom.



MCU: ESP32 V1 → ESP32-C3 RISC-V

4 Sensor Node

Initial choice

- ESP32 DOIT DevKit V1 dual-core
- Higher idle current: ~ 4 mA (deep sleep)
- Oversized for the task

Result

- ESP32-C3
- $50 \mu\text{A}$ deep sleep
- 20 mA active

Deep-sleep: ESP32-C3 $50 \mu\text{A}$ vs. ESP32 DOIT ~ 4 mA; $80\times$ reduction in quiescent current



Protocol: BLE $\rightarrow$ ESP-NOW

5 Edge & Gateway

Initial choice

- BLE for local comms
- ~ 500 ms pairing
- ~ 7.5 ms TX latency

Result

- ESP-NOW
- **No pairing overhead**
- < 2 ms TX

ESP-NOW radio-on: < 2 ms, no association step vs. BLE pairing ~ 500 ms + 7.5 ms TX



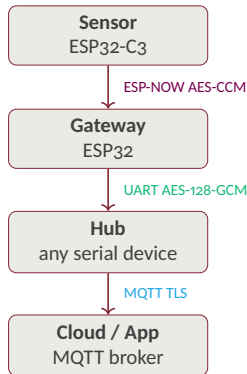
Three-Tier Architecture: Why a Dedicated Gateway?

5 Edge & Gateway

- **ESP-NOW stays local:** sensor never touches Wi-Fi; no energy cost of association or scanning when router changes or reboots
- **Hub is transport-agnostic:** gateway exports UART + AES-128-GCM; hub can be Raspberry Pi, laptop, or cloud VM; not limited to another ESP32
- **Network isolation:** ESP-NOW subnet is air-gapped from the internet; gateway is the single trusted boundary between sensor network and IP world

Isolation property

Sensor carries no IP stack and no internet credentials; PMK/LMK scope is limited to the local ESP-NOW subnet.





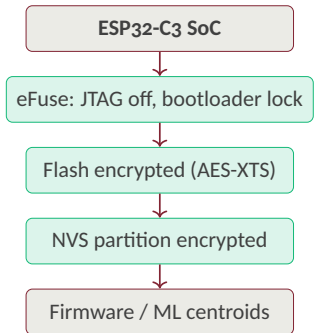
Hardware Security Foundation

6 Security

ESP-IDF Framework

All firmware uses **ESP-IDF** (not Arduino) for direct hardware-level security API access.

- **Encrypted Flash (AES-XTS):** entire firmware image on NOR flash encrypted at rest; plaintext never leaves the chip
- **Encrypted NVS partition:** crypto keys and ML model centroids stored in a dedicated NVS region with separate AES encryption
- **eFuse write-once protection:** JTAG debug port disabled, bootloader locked; survives firmware reflash; root of trust for the device



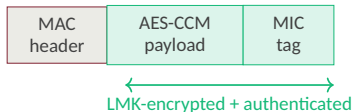


Link Security: ESP-NOW (Sensor → Gateway)

6 Security

- **Algorithm:** AES-CCM (CCMP per 802.11); handled by Wi-Fi hardware accelerator (< 0.1 ms overhead)
- **PMK** (Primary Master Key): 16-byte key that initialises the ESP-NOW security subsystem; used to derive session material
- **LMK** (Local Master Key): 16-byte key unique per registered peer (`.encrypt = true`); all sensor ↔ gateway frames encrypted with this LMK
- Keys stored in **encrypted NVS** on both devices

ESP-NOW Secured Frame



PMK/LMK hierarchy

A compromised sensor leaks only its LMK, not the master key used to initialise other peers.

Sensor
ESP32-C3

Band 1

ESP-NOW +
AES

PMK (network key) +
per-peer LMK (16 bytes each)
Hardware AES on
ESP32 chip

Gateway
ESP32-C3

Band 2

UART +
AES-128-GCM

12-byte random IV per
frame (HW RNG)
16-byte auth tag
Key from encrypted NVS

Edge Hub
ESP32

Band 3

MQTTs +
TLS 1.2

Self-signed CA
Embedded cert in
firmware
Port 8883

Cloud
Mosquitto



Keys generated by ESP32
hardware RNG on first boot



Stored in encrypted NVS
(nvs_keys partition)



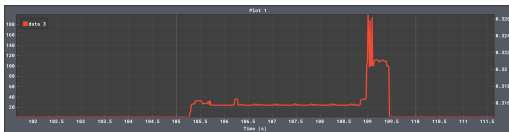
Each hop independently
authenticated



Window Energy: Alarm vs. No Alarm

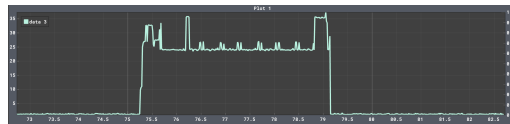
7 System Analysis

Alarm Detected



Vibration profile during an active intrusion attempt. The energy rapidly escalates past 200, breaching the Dynamic Cluster Threshold and triggering an alarm.

No Alarm Detected



Normal environmental vibrations (wind, raindrops). Signal peaks around 35, safely remaining within baseline boundaries and preventing false triggers.



Battery Life vs. Duty Cycle (3500 mAh)

7 System Analysis

D (%)	Active/hr	I_{mean}	Battery life
0.01	0.4 s	52 μ A	7.7 years
0.1	3.6 s	70 μ A	5.7 years
0.5	18 s	150 μ A	2.7 years
1.0	36 s	250 μA	1.6 years
5.0	3 min	1.05 mA	4.6 months
10.0	6 min	2.05 mA	2.3 months
20.0	12 min	4.04 mA	1.2 months
50.0	30 min	10.0 mA	14.5 days

$$I_{mean} = D \cdot 20 \text{ mA} + (1 - D) \cdot 50 \mu\text{A}$$

$D = 1\%$: operating point

36 s active per 3600 s; ADXL362 INT1 gates all MCU wake events.

- ADXL362 activity threshold: MCU wakes only when acceleration exceeds configured mg limit
- ESP-NOW TX: 1.5 ms at ~ 200 mA peak; contribution < 0.01 mA to I_{avg}
- $t_{life} \approx 1.6$ years on 3500 mAh



Battery Life vs. Duty Cycle (3500 mAh) DEVIATION

7 System Analysis

D (%)	Active/hr	I_{mean}	Battery life
0.01	0.4 s	55 μ A	7.3 years
0.1	3.6 s	100 μ A	4.0 years
0.5	18 s	300 μ A	1.3 years
1.0	36 s	550 μA	8.7 months
5.0	3 min	2.55 mA	1.9 months
10.0	6 min	5.05 mA	29 days
20.0	12 min	10.0 mA	14.5 days
50.0	30 min	25.0 mA	5.8 days

$$I_{active} = \frac{2.5 \text{ s} \cdot 20 \text{ mA} + 0.5 \text{ s} \cdot 200 \text{ mA}}{3.0 \text{ s}} = 50 \text{ mA}$$

$D = 1\%$: Worst-case deviation

Active time per alarm: 3.0 s
(2.5 s sampling/FFT + 0.5 s Wi-Fi retry peak)

- **Active current deviation:** rises to 50 mA average due to prolonged 200 mA TX retries
- **Significant reduction:** t_{life} drops from 1.6 years to **8.7 months**
- Highlights the importance of low-power ESP-NOW over regular Wi-Fi

$$I_{mean} = D \cdot I_{active} + (1 - D) \cdot 50 \mu\text{A}$$



Detection Latency: End to End

7 System Analysis

Stage	Duration	Cumulative
ADXL362 INT1 fires (hardware threshold)	< 1 ms	< 1 ms
ESP32-C3 wakes from deep sleep	~ 10 ms	~ 11 ms
Sample 512 points @ 200 Hz	2560 ms	2571 ms
FFT + 51-feature extraction	< 50 ms	2621 ms
K-means inference ($K = 10$)	< 10 ms	2631 ms
ESP-NOW TX to gateway	~ 1.5 ms	2633 ms
UART AES-GCM bridge to hub	< 5 ms	2638 ms
MQTT publish to cloud	< 100 ms	< 2.74 s

Key insight

2.56 s acquisition window: $\Delta f = 200/512 \approx 0.39$ Hz resolution; covers sub-100 Hz window vibration band. Classification adds < 200 ms.



Possible Improvements

8 Improvements

- **ESP-Provisioning**

Integrate ESP-Provisioning to facilitate seamless and secure Wi-Fi and cloud pairing of the hub.

- **Continuous ML Optimization**

Periodically acquire new environmental vibration data to retrain and dynamically optimize the ML model.

- **Mesh Networking**

Deploy a mesh network among gateways to extend communication range and improve system redundancy.

- **Energy Harvesting**

Integrate small solar panels to achieve fully self-sustaining sensor node operation and indefinite battery life.



KnockKnock *Thank you for listening!*

Any questions?